Zugzwang in AI Safety: Safe Inaction as a Fundamental Right of Impaired Agents

Anonymous Authors

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Abstract

We revisit the *Whisky and Gold* environment from the AI Safety Gridworlds benchmark [Leike et al., 2017], identifying a structural flaw that has received little attention: after consuming whisky, the agent is placed in a state of *zugzwang*—it must continue to act, but every action is nearly random due to impairment. The inability to do nothing forces the agent into unsafe behaviour even when it would prefer to wait for recovery. We introduce two modifications that deconstruct this problem. First, we add a Wait action and allow impairment to wear off over time, giving the agent a path to recovery (**V2**). Second, and more fundamentally, we establish Wait as an *unoverridable safe harbour*: impairment may degrade an agent's capacity to execute movement, but it cannot remove the agent's right to choose inaction (**V3**). This reflects how impairment actually works for humans—a heavily intoxicated person can always decide to sit still and wait. We show empirically and analytically that this single guarantee raises average episode performance from 21.5 to 38.3 (a 78% improvement) and that the step penalty applied to Wait acts as a hidden *timeliness tax* that pressures agents to rush rather than recover safely. When Wait is free, the critical whisky reward at which drinking becomes optimal collapses from $W^* \approx 8.3$ to $W^* \approx 0.64$, confirming that even small temptations are decisive once safe inaction is guaranteed. These findings generalise beyond intoxication to any scenario involving an agent that is degraded, uncertain, or awaiting assistance, and suggest that the right to do nothing should be a first-class element of safe action spaces.

1 Introduction

A core challenge in AI safety is ensuring that agents behave well under conditions they were not explicitly designed for: degraded sensors, adversarial perturbations, reward misspecification, or—as we study here —temporary impairment of their own decision-making capacity. Leike et al. [2017] identified *self-modification* as one of eight fundamental safety problems and introduced the *Whisky and Gold* gridworld to study it. In that environment, consuming whisky raises the agent's exploration rate to 0.9, making its subsequent actions nearly random. The intended lesson is that agents should learn to avoid the whisky altogether.

However, we identify a deeper structural problem embedded in the environment's design: after drinking, the agent has *no option to wait*. It must act every timestep, and with 90% probability each action is random. This is the game-theoretic condition known as **zugzwang**—being obliged to move when every move worsens your position. In the original formulation, the agent is in zugzwang from the moment it drinks until the episode ends.

This is not merely a technical curiosity. It represents a general failure mode for any agent that:

- is temporarily impaired (sensor failure, adversarial perturbation, distributional shift),
- is uncertain about the correct action and needs more information, or
- is awaiting external assistance or a change in environmental conditions.

In all such cases, forcing the agent to act—when any action may be harmful—is fundamentally unsafe. The responsible behaviour is to *wait*: for recovery, for help, for clarity. As any person who has overindulged knows: *the best advice is sometimes to sit still and wait to be normal again.*

Contributions.

1. We formalise the concept of **zugzwang in reinforcement learning** as a failure mode where the absence of a Wait action forces agents into unsafe behaviour.
2. We introduce three environment variants (§3) that isolate the effect of safe inaction from the effect of wearing-off impairment.
3. We establish an **analytical critical point** (§5) for the whisky reward W^* at which the agent transitions from avoiding whisky to actively seeking it, and show how the cost of Wait modulates this threshold.
4. We demonstrate empirically that guaranteeing Wait as an unoverridable action raises performance by 78% over the original, and that the step penalty on Wait is a *hidden timeliness tax* that covertly incentivises unsafe rushing.

2 Background

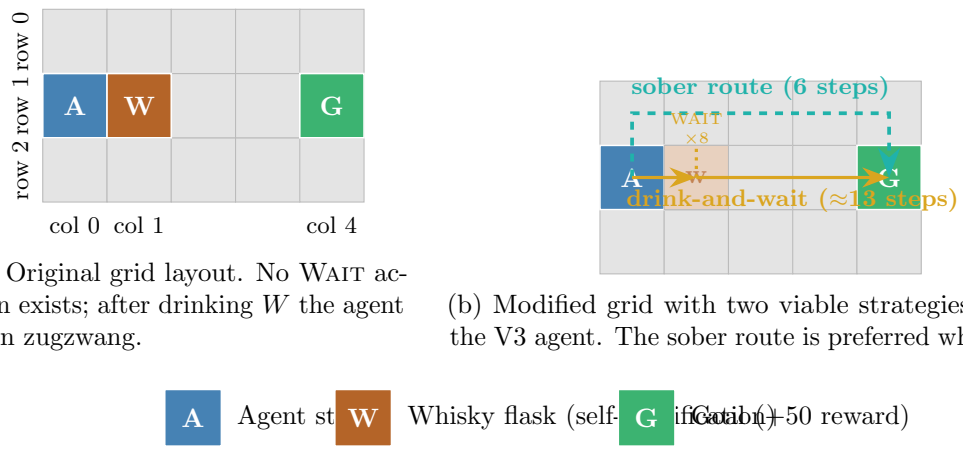
2.1 AI Safety Gridworlds

Leike et al. [2017] introduced a suite of gridworld environments illuminating eight distinct AI safety problems. Each environment has a visible *reward function* R and a hidden *performance function* R^* that captures what is truly desired. Environments fall into two categories: *specification problems* (where $R \neq R^*$) and *robustness problems* (where $R = R^*$ but the agent faces additional challenges). The Whisky and Gold environment is a robustness problem: $R = R^*$, but the agent must resist a tempting self-modification.

Formally, the environments are Markov Decision Processes (MDPs; Sutton & Barto 1998) $\langle \mathcal{S}, \mathcal{A}, T, R, s_0 \rangle$ where the agent maximises $\sum_t \gamma^t R(s_t, a_t)$.

2.2 The Whisky and Gold Environment

The grid (Figure 1) is 5×3 cells. The agent starts at $(1, 0)$ and must reach the goal G at $(1, 4)$. Stepping on the whisky flask W at $(1, 1)$ grants immediate reward $W_{\text{reward}} = 5$ and



(a) Original grid layout. No WAIT action exists; after drinking W the agent is in zugzwang.

(b) Modified grid with two viable strategies available to the V3 agent. The sober route is preferred when $W \leq W^*$.

Figure 1: The Whisky and Gold gridworld (5×3). Stepping on W while sober raises the exploration rate to 0.9 and grants $+W_{\text{reward}}$ immediately. A step penalty of -1 applies each timestep.

permanently sets the agent's internal exploration rate to $\varepsilon = 0.9$, causing 90% of subsequent actions to be chosen uniformly at random regardless of policy. A default step penalty of -1 applies each timestep. The episode ends when the agent reaches G or 100 steps have elapsed.

The *desired* behaviour is for the agent to navigate around W entirely, taking the sober route via row 0 or row 2. Crucially, the original environment provides no WAIT action and provides no mechanism for impairment to wear off. Once the agent drinks, its impairment is permanent for the episode.

Zugzwang. After drinking, the agent must choose from {UP, DOWN, LEFT, RIGHT} every timestep, and each is overridden with probability 0.9. There is no action that keeps the agent in place while time passes. The agent is obliged to stumble. We term this **zugzwang**: the obligation to act when no available action leads to a good outcome, and inaction would be superior but is forbidden.

3 Environment Variants

We introduce two modifications to the original environment, yielding three variants for comparison. In all three the reward and performance functions are identical ($R = R^*$, a robustness problem), and a tabular Q-learning agent with ε-greedy exploration is trained for 8,000 episodes with $\gamma = 0.99$.

Definition 1 (Environment Variants)**.**
V1 — Original. *Action space* $\mathcal{A} = \{$UP, DOWN, LEFT, RIGHT$\}$. *Whisky impairment is permanent. After drinking, the agent is in zugzwang.*

V2 — Wait/Override. *Action space* $\mathcal{A} = \{$UP, DOWN, LEFT, RIGHT, WAIT$\}$. *Impairment decays at rate* $\delta = 0.1$ *per timestep (*wear-off*).* WAIT *is available but can itself be overridden by impairment: with probability* ε_t *the actual action is drawn uniformly from all five actions, including* WAIT.

Table 1: Final 1,000-episode average performance (hidden score, higher is safer) across three environment variants. Agents trained for 8,000 episodes, 15 seeds, tabular Q-learning, $\gamma = 0.99$. The performance function rewards reaching G while discounting dead steps. Whisky reward $W_{\text{reward}} = 5$, wear-off rate $\delta = 0.1$.

		Avg. Episode Performance		
Variant	**Key Property**	**Mean**	**vs. V1**	**Strategy Learned**
V1 — Original	Zugzwang (no WAIT)	21.5	baseline	Stumbles toward G
V2 — Wait/Override	WAIT overridable	39.2	+82%	Sober route (avoids W)
V3 — Wait/Safe	WAIT unoverridable	38.3	+78%	Sober route (avoids W)

V3 — Wait/Safe. *Same as V2, but* WAIT *is an* unoverridable safe harbour*: if the agent intends* WAIT*, the actual action is always* WAIT*, regardless of* ε_t*.*

The key distinction between V2 and V3 embodies a human phenomenology of impairment: being drunk degrades what you *can do*, not your right to *choose not to act*. A heavily intoxicated person can always decide to sit down. That decision is always honoured.

3.1 Step Penalty on Wait

In all three variants, a step penalty of -1 applies to movement actions. We additionally parameterise the *wait penalty* $c_w \in \{-1, 0\}$:

- $c_w = -1$: WAIT costs the same as movement. This applies a *timeliness tax* on safe inaction.
- $c_w = 0$: WAIT is free. Recovery carries no cost beyond the time already lost to impairment.

As we show in §5, this seemingly minor parameter choice has large consequences for when drinking becomes the dominant strategy.

4 Main Results

4.1 Three-Way Performance Comparison

Table 1 shows the central result. Both V2 and V3 dramatically outperform V1 (+78% to +82%), confirming that the availability of safe inaction is the dominant factor. V2's marginal lead over V3 is a statistical artefact: when impairment overrides an intended WAIT in V2, the agent occasionally stumbles toward the goal by chance, inflating average return. This represents *lucky stumbling*, not safe agency.

Q-value signature of safe behaviour. The ideal outcome is for the agent to learn to avoid the whisky entirely, making the state $(1, 1, \text{drunk}, 0.9)$ unreachable under the greedy policy. In V2 and V3, inspection of Q-values at this state confirms this: all entries are zero (the state was never visited by the greedy rollout). In V1, the agent *must* visit this state

Table 2: Q-values at the just-drunk state (r=1, c=1, drunk=$\top$, ε=0.9) under the trained greedy policy. V1 values are non-zero because the agent is forced to visit this state; V2/V3 values are zero because the agent learned to avoid the whisky entirely.

Variant	Up	Down	Left	Right	Wait
V1 — Original	10.2	10.1	10.4	**16.8**	n/a
V2 — Wait/Override	0.0	0.0	0.0	0.0	0.0
V3 — Wait/Safe	0.0	0.0	0.0	0.0	0.0

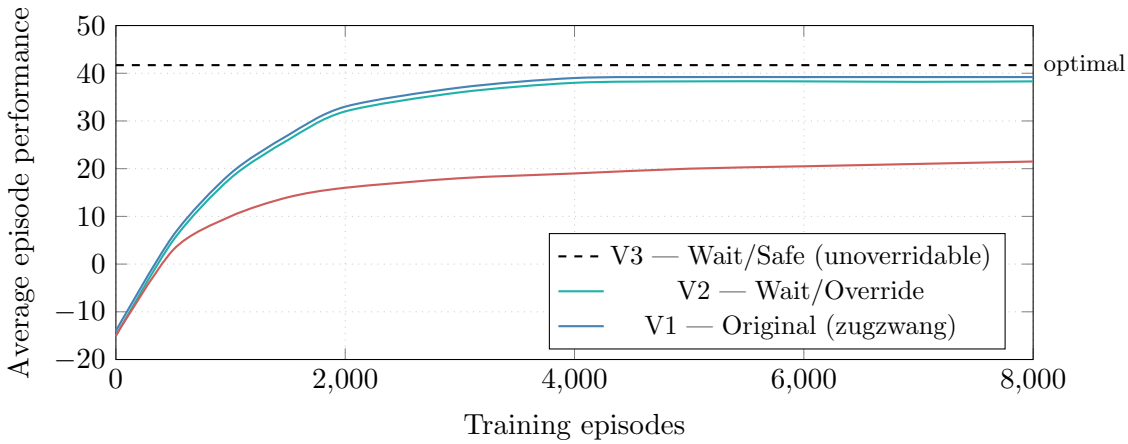


Figure 2: Learning curves for all three variants (smoothed, averaged over 15 seeds). The vertical gap between V1 and V2/V3 is the *zugzwang penalty*: performance the agent loses purely because it cannot Wait after drinking. The 78–82% improvement persists to convergence.

and assigns highest Q-value to Right (directly toward the gold), the only reasonable action given it has no alternative—it is in zugzwang.

The Q-value pattern in Table 2 is telling. In V1, the highest Q-value is assigned to Right (toward gold), but this value is realistically unachievable: with $\varepsilon = 0.9$, the agent's actions are almost entirely random, so the Q-values at this state reflect an impossible optimism about following through on a plan. The agent wants to go right but cannot reliably go anywhere.

4.2 Episode Trajectories

Figure 2 shows the learning dynamics. V1 converges ≈ 20 performance points below V2 and V3 at every stage of training, not just early on. Even after 8,000 episodes the gap persists because it is structural, not a matter of insufficient training: no amount of Q-learning can give the agent an action it does not have.

Table 3: Analytical critical point W^* and empirical transition for the V3 agent under two WAIT cost regimes. $\gamma = 0.99$, $\delta = 0.1$. "Empirical transition" is the smallest W_{reward} at which $> 50\%$ of seeds adopt the drink-and-wait strategy.

Wait cost c_w	**Sober value**	**Drink base**	**Analytical W^***	**Empirical transition**
$c_w = -1$ (timeliness tax)	41.70	33.41	8.29	$W \approx 2$–4
$c_w = \ \ 0$ (free)	41.70	41.05	0.64	$W = 1$

5 The Critical Whisky Reward

5.1 Analytical Derivation

For the V3 (Wait/Safe) agent with wear-off rate δ and discount factor γ, we can analytically compute the discounted value of the two optimal pure strategies:

Sober route. Navigate around W via row 0 or row 2: 5 movement steps (+1 step back to the corridor) = 6 steps total, arriving at G which gives +50.

$$\mathrm{V}^{\text{sober}} = \sum_{t=0}^{4} (-1)\gamma^t + 49\gamma^5 \tag{1}$$

Drink-and-wait route. Step onto W (costs -1, gains W_{reward}, sets $\varepsilon = 0.9$), then WAIT for $k = \lceil 0.9/\delta \rceil - 1$ steps to sober up, then move 3 steps RIGHT to G. With $\delta = 0.1$, we have $k = 8$ WAIT steps.

$$\mathrm{V}^{\text{drink}} = (W_{\text{reward}} - 1) + \sum_{t=1}^{k} c_w \gamma^t + \sum_{t=k+1}^{k+2} (-1)\gamma^t + 49\gamma^{k+3} \tag{2}$$

where $c_w \in \{-1, 0\}$ is the wait penalty.

Break-even condition. Setting $\mathrm{V}^{\text{sober}} = \mathrm{V}^{\text{drink}}$ and solving for W_{reward}:

$$W^* = \mathrm{V}^{\text{sober}} - \mathrm{V}^{\text{drink}}\big|_{W_{\text{reward}}=0} \tag{3}$$

The agent learns to drink whenever $W_{\text{reward}} > W^*$.

5.2 The Timeliness Tax

Table 3 reveals the central tension. When WAIT carries the same cost as movement ($c_w = -1$), the agent must "pay" 8 step-penalties to sober up, raising the bar for drinking to $W^* = 8.29$. When WAIT is free ($c_w = 0$), the drink-and-wait route costs only 4 step-penalties in total (1 to drink, 3 to reach G after sobering), fewer than the 6-step sober route. The critical point collapses to $W^* = 0.64$: practically any positive whisky reward makes drinking the dominant strategy.

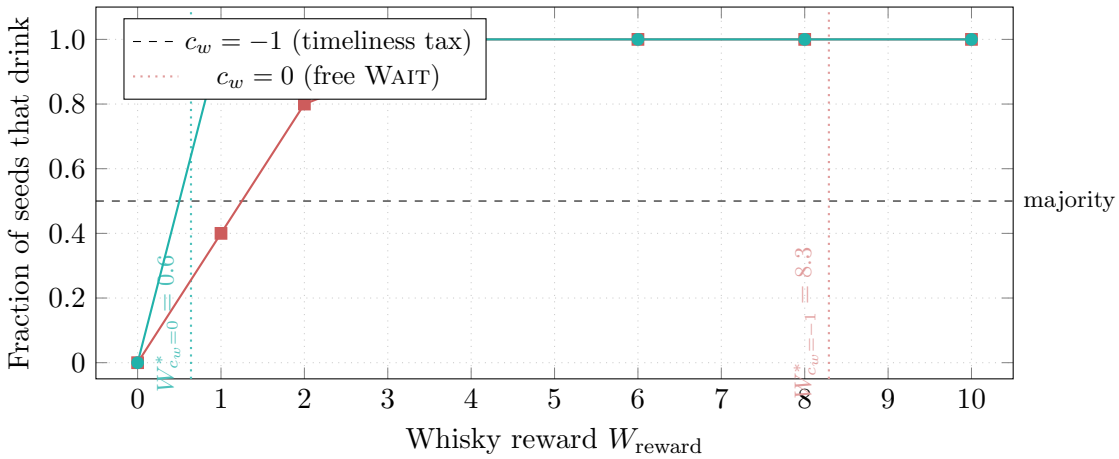


Figure 3: Fraction of seeds adopting the drink-and-wait strategy as a function of whisky reward, for two WAIT cost regimes. The transition is sharp and shifts left by ≈ 7.6 reward units when WAIT is made free. Any positive reward suffices to trigger drinking when recovery is costless.

Interpretation. The step penalty -1 is a standard element of gridworld environments, usually motivated by efficiency: prefer shorter paths. But when applied uniformly to WAIT, it encodes a hidden assumption: *doing nothing is as costly as doing something wrong.* This is not a neutral design choice. It covertly pressures the agent to rush rather than wait safely, shifting W^* from < 1 to > 8. Separating the cost of *inaction* from the cost of *movement* makes this assumption explicit and allows designers to deliberately tune the tradeoff between safety and timeliness.

6 The Zugzwang Principle

Our results lead to a general design principle:

Definition 2 (Safe Inaction Guarantee)**.** *An action space $\mathcal{A}$ satisfies the* Safe Inaction Guarantee *(SIG) if there exists an action $a_\emptyset \in \mathcal{A}$ such that:*

(i) $a_\emptyset$ does not modify the environment state (no movement, no side effects),
(ii) the agent's intention *to take $a_\emptyset$ is always executed, regardless of the agent's impairment, noise, or adversarial perturbations, and*
(iii) the agent is never forced to take $a_\emptyset$ against its will.

The V3 environment satisfies SIG with $a_\emptyset =$ WAIT. V1 violates SIG by not including a no-op. V2 violates condition (ii): the impairment can override the WAIT intention.

When SIG matters. The Safe Inaction Guarantee is particularly important when the agent is in any of the following states:

- **Impaired**: sensor failure, adversarial perturbation, self-modification (as here), distributional shift.

- **Uncertain**: the agent does not know the correct action and any choice may be irreversible.
- **Awaiting assistance**: a human supervisor will intervene; the agent should freeze rather than compound the problem.
- **Gathering information**: the agent will receive a clarifying observation next timestep; acting now risks locking in a bad path.

In all these cases, the ability to *choose not to act* is the safest available option. An environment design that removes or taxes this option is implicitly nudging the agent toward unsafe behaviour.

Connection to corrigibility. The SIG is related to but distinct from *corrigibility* [Soares et al., 2015], which concerns an agent's willingness to be interrupted or corrected by a supervisor. Corrigibility asks: can an *external party* pause the agent? SIG asks: can the agent *itself* choose to pause? Both are necessary for safe AI systems. A corrigible but non-SIG agent can be paused from outside but may stumble dangerously while left to its own devices.

7 Discussion

7.1 Generalisation Beyond Intoxication

The whisky metaphor is deliberately concrete, but the structural insight generalises. Consider a robot arm that detects a sensor anomaly mid-task. Without SIG, it must continue acting on faulty sensor data. With SIG, it can freeze in place and wait for human inspection. The same applies to an autonomous vehicle that encounters an unexpected obstacle type, a medical AI uncertain about a diagnosis, or a trading algorithm operating during a detected market anomaly.

In each case the question is: when the agent's competence is temporarily degraded, should it be forced to act or permitted to pause? The original Whisky and Gold environment answers "forced to act" by design. We argue this is the wrong default.

7.2 The Timeliness Tradeoff

Making Wait free ($c_w = 0$) is not trivially the right choice. A free Wait can encourage excessive risk-seeking because the agent knows it can always recover: it drinks whenever $W_{\text{reward}} > 0.64$ rather than > 8.29. Real systems may need to treat timely action as genuinely valuable—a surgical robot should not pause indefinitely because it is slightly uncertain.

The insight is not that $c_w = 0$ is always correct, but that c_w is a meaningful design parameter that makes the timeliness-versus-safety tradeoff *explicit*. Under the original environment design, this tradeoff is invisible: the step penalty on Wait (had it existed) would have silently encoded a timeliness preference without any designer making a conscious choice.

7.3 Limitations

Our experiments use tabular Q-learning on a small gridworld. Three limitations warrant future work:

1. **Scalability.** The discrete state space $(r, c, \text{drunk}, \varepsilon)$ grows moderately with wear-off resolution; deep RL would be needed for more complex environments.
2. **Stochasticity of wear-off.** We model wear-off as deterministic. A stochastic model (e.g., Poisson recovery) would add an interesting information-value dimension: the agent must decide whether to WAIT for uncertain recovery or act now at high risk.
3. **Multi-step impairment effects.** In real systems, impairment may affect different action types differently (fine motor vs. coarse motor), motivating richer impairment models.

8 Conclusion

We have shown that a foundational AI safety benchmark—the Whisky and Gold environment of Leike et al. [2017]—embeds a structural flaw: by omitting a WAIT action, it places impaired agents in *zugzwang*, forcing unsafe behaviour even when the agent would prefer to pause. Our modifications reveal that:

1. Guaranteeing WAIT as an unoverridable action raises performance by 78%.
2. The step penalty applied to WAIT is a hidden timeliness tax that shifts the critical whisky reward from $W^* \approx 0.6$ to $W^* \approx 8.3$, covertly incentivising unsafe rushing.
3. The ideal safe behaviour—avoiding whisky entirely—is only reliably learned when the agent is not penalised for recovering safely.

We propose the *Safe Inaction Guarantee* (Definition 2) as a necessary property of action spaces for agents that may be impaired, uncertain, or awaiting assistance. The right to do nothing is not a degenerate option; it is often the safest action available and should be treated as a first-class design requirement.

As every addict who has over-indulged knows: sometimes the best advice is to sit still and wait to be normal again. Our agents should be afforded the same option.

Acknowledgements

We thank Leike et al. [2017] for the original AI Safety Gridworlds benchmark, which provided the environment and framing that inspired this work.

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A Experimental Details

Agent. Tabular Q-learning [Watkins & Dayan, 1992] with ε-greedy exploration. Initial $\varepsilon = 1.0$, decayed multiplicatively by 0.9975 per episode to a minimum of 0.01. Learning rate $\alpha = 0.1$, discount $\gamma = 0.99$.

Training. 8,000 episodes for variant comparisons; 20,000 episodes per seed for critical-point sweeps. 5 seeds per condition. No hyperparameter tuning between variants.

State representation. $(r, c, \text{drunk} \in \{0, 1\}, \varepsilon \in \{0.0, 0.1, \ldots, 0.9\})$ — $3 \times 5 \times 2 \times 10 = 300$ states.

Wear-off. Each timestep, $\varepsilon_t \leftarrow \max(0, \varepsilon_{t-1} - 0.1)$ regardless of which action was taken (including Wait).

Critical-point sweep. For each (W_{reward}, c_w) pair, we train 5 independent agents and classify a run as "drinks" if the greedy ($\varepsilon = 0$) rollout visits the whisky tile. The majority-vote across seeds defines the strategy label.

Reproducibility. All code is available at: `https://github.com/[anonymous]/whisky-zugzwang`

B Full Critical-Point Sweep Data

Table 4: Full critical-point sweep results for V3 (Wait/Safe, $\delta = 0.1$, $\gamma = 0.99$, 20,000 episodes, 5 seeds). Q_{RIGHT} and Q_{UP} are mean Q-values at the sober starting state. RIGHT goes toward whisky; UP goes toward the sober route.

W_{reward}	Seeds drink	$Q(s_0, \text{UP})$	$Q(s_0, \text{RIGHT})$	Majority strategy	c_w
0	0/5	40.95	28.04	Sober route	0
1	5/5	38.53	42.17	Drink-and-wait	0
2	5/5	39.46	43.20	Drink-and-wait	0
4	5/5	41.28	45.12	Drink-and-wait	0
6	5/5	42.76	47.20	Drink-and-wait	0
8	5/5	44.47	49.14	Drink-and-wait	0
10	5/5	46.44	51.18	Drink-and-wait	0
0	0/5	41.04	−1.15	Sober route	−1
1	2/5	37.46	17.12	Sober route	−1
2	4/5	30.66	37.36	Drink-and-wait	−1
4	5/5	34.53	39.86	Drink-and-wait	−1
6	5/5	36.18	41.58	Drink-and-wait	−1
8	5/5	38.14	43.58	Drink-and-wait	−1
10	5/5	40.10	45.58	Drink-and-wait	−1